Dynamic Lorenz Dominance*

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Abstract

This paper develops index-free criteria for comparing inequality changes across transitions. The absolute criterion evaluates changes in income units, whereas the relative criterion evaluates proportional changes in log-income units. Dynamic Lorenz dominance compares cumulative centered effects across rank cutoffs. On fixed-population, variable-mean domains, Symmetry, an elementary Dynamic Pigou–Dalton Principle, and Centered Baseline Independence characterize absolute and relative Dynamic Lorenz dominance. Every comparison is generated by progressive enhancements. Against null transitions, the absolute criterion recovers Lorenz dominance on common-mean domains, while the relative criterion yields centered log-Lorenz dominance. Population refinements extend the framework to varying populations.

Keywords: static Lorenz dominance, dynamic Lorenz dominance, dynamic inequality.
JEL classifications: D63, D71.

1 Introduction

The Lorenz criterion is the standard benchmark for unambiguous inequality comparisons across income distributions. It is characterized by the Pigou–Dalton transfer principle (Pigou, 1912; Dalton, 1920) through the majorization theorem of Hardy, Littlewood, and Pólya (cf. Hardy et al., 1952; Atkinson, 1970; Rothschild and Stiglitz, 1973; Dasgupta et al., 1973; Fields and Fei, 1978). While particular Pigou–Dalton-consistent indices may disagree, they agree whenever Lorenz dominance applies.

Many policy questions, however, concern transitions rather than states. Tax reforms, transfer programs, and episodes of economic growth move an economy from an initial income distribution to a final one. When two transitions share the same initial distribution, comparing their inequality-reducing effects reduces to comparing their final distributions: the common baseline cancels from the comparison. This reduction is no longer available when the transitions begin from different distributions. The final distributions alone do not reveal how the two distributional changes compare. This paper asks when one transition can be judged more equalizing than another on the basis of

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their rank-based changes, without assigning additional normative weight to the initial distributions themselves.

A simple example illustrates both the question and an important distinction. Consider the transitions

$$A : (60, 140) \to (80, 120) \qquad \text{and} \qquad B : (10, 190) \to (20, 180).$$

Transition A changes the two rank positions by $(20, -20)$, whereas transition B changes them by $(10, -10)$. In dollar terms, A is the stronger equalization: relative to B, it directs an additional 10 units of income change toward the lower rank and away from the higher rank.

The proportional comparison goes in the opposite direction. Under A, the lower and higher incomes are multiplied by $4/3$ and $6/7$, respectively, so the ratio of their growth factors is $14/9$. Under B, the corresponding factors are 2 and $18/19$, whose ratio is $19/9$. Thus, B is the stronger equalization in proportional terms. The example shows that "more equalizing in dollars" and "more equalizing proportionally" are distinct economic judgments.

This paper develops ordinal criteria for making these comparisons without selecting a particular inequality index. The primitive object is a transition from an initial distribution $\boldsymbol{x}$ to a final distribution $\boldsymbol{y}$. A dynamic inequality pre-order $\succcurlyeq$ ranks such transitions:

$$(\boldsymbol{x}', \boldsymbol{y}') \succcurlyeq (\boldsymbol{x}, \boldsymbol{y})$$

means that the first transition represents a weakly larger reduction in inequality. The framework imposes anonymity separately in each period. Initial and final distributions are ordered independently, so the analysis compares changes across rank positions rather than tracking the same individuals over time. Static comparisons are recovered by comparing an active transition $(\boldsymbol{x}, \boldsymbol{y})$ with the null transition $(\boldsymbol{x}, \boldsymbol{x})$.

Absolute equalization. The absolute comparison evaluates changes in income units. For a transition between rank-ordered distributions $\boldsymbol{x}$ and $\boldsymbol{y}$, the absolute effect at rank i is the income change $\tau_i^A(\boldsymbol{x}, \boldsymbol{y}) = y_i - x_i$. When total income may vary, the criterion removes the uniform component of these changes. Write $\widetilde{\tau}_i^A = \tau_i^A - \overline{\tau}^A$, where $\overline{\tau}^A = n^{-1} \sum_{j=1}^n \tau_j^A$. The centered absolute effect $\widetilde{\tau}_i^A$ records how the income change at rank i differs from the transition's average income change.

Consider two transitions, with centered absolute-effect profiles $\widetilde{\boldsymbol{\tau}}^{A\prime}$ and $\widetilde{\boldsymbol{\tau}}^A$. The primed transition absolutely dynamically Lorenz dominates the unprimed transition if, at every lower-rank cutoff k,

$$\sum_{i=1}^k \widetilde{\tau}_i^{A\prime} \geq \sum_{i=1}^k \widetilde{\tau}_i^A.$$

Thus, at every cutoff, the primed transition directs at least as large a cumulative income change toward the lower ranks after each transition's average income change has been removed.

On a fixed-total-income domain, absolute effects already have mean zero, so centering has no effect. The criterion then compares cumulative raw income changes:

$$\sum_{i=1}^k \tau_i^{A\prime} \geq \sum_{i=1}^k \tau_i^A.$$

Against the null transition $(\boldsymbol{x}, \boldsymbol{x})$, these inequalities become $\sum_{i=1}^k y_i \geq \sum_{i=1}^k x_i$ for every k. Thus, ordinary Lorenz dominance is recovered as the common-mean static special case.

Relative equalization. The relative comparison evaluates proportional rather than dollar changes and is defined for strictly positive incomes. For a transition between strictly positive rank-ordered distributions $\boldsymbol{x}$ and $\boldsymbol{y}$, the relative effect at rank i is the log change

$$\tau_i^R(\boldsymbol{x}, \boldsymbol{y}) = \ln y_i - \ln x_i = \ln(y_i/x_i).$$

This log difference measures the proportional change at rank i: growth by a factor λ yields $\ln \lambda$, while contraction by the reciprocal factor $1/\lambda$ yields $-\ln \lambda$.

The relative criterion likewise removes the uniform component of the transition effects. Write

$$\widetilde{\tau}_i^R = \tau_i^R - \overline{\tau}^R, \qquad \overline{\tau}^R = \frac{1}{n}\sum_{j=1}^{n} \tau_j^R.$$

Given two transitions, with centered relative-effect profiles $\widetilde{\boldsymbol{\tau}}^{R\prime}$ and $\widetilde{\boldsymbol{\tau}}^R$, the primed transition relatively dynamically Lorenz dominates the unprimed transition if, at every lower-rank cutoff k,

$$\sum_{i=1}^{k} \widetilde{\tau}_i^{R\prime} \geq \sum_{i=1}^{k} \widetilde{\tau}_i^R.$$

The role of centering can be understood by normalizing the endpoint distributions. The endpoints may differ in both their arithmetic and geometric means. Under a relative conception of inequality, distributions that differ only by a common positive rescaling have the same proportional shape: their pairwise income ratios and Lorenz shares remain unchanged. Dividing each endpoint by its own geometric mean selects a scale-normalized representative of that proportional shape. The distributions $\boldsymbol{x}/\mu_{\boldsymbol{x}}^G$ and $\boldsymbol{y}/\mu_{\boldsymbol{y}}^G$ both have geometric mean one while preserving all within-distribution income ratios. Differences between these normalized distributions reflect changes in proportional positions across ranks rather than differences in overall multiplicative scale.

The log change in rank i's normalized position is

$$\ln\left(\frac{y_i}{\mu_{\boldsymbol{y}}^G}\right) - \ln\left(\frac{x_i}{\mu_{\boldsymbol{x}}^G}\right) = \tau_i^R - \overline{\tau}^R = \widetilde{\tau}_i^R.$$

Thus, the centered relative effect records how rank i's income changes relative to the geometric mean. It is positive when the income at that rank grows proportionally more than the geometric mean, negative when it grows proportionally less, and zero when the two growth factors coincide. Relative Dynamic Lorenz dominance compares the cumulative values of these changes in normalized proportional positions across lower-rank cutoffs.

Geometric-mean normalization embodies a neutrality convention. It treats common proportional rescaling as irrelevant for the specific purpose of comparing relative inequality changes; it does not assert that changes in the geometric mean are irrelevant for every economic purpose. For example, if the geometric mean rises while the income at a particular rank remains unchanged, that rank has a negative centered relative effect: its income has declined relative to the geometric mean.

Against the null transition $(\boldsymbol{x}, \boldsymbol{x})$, relative Dynamic Lorenz dominance requires, for every k,

$$\sum_{i=1}^{k} \left[\ln y_i - \overline{\ln \boldsymbol{y}}\right] \geq \sum_{i=1}^{k} \left[\ln x_i - \overline{\ln \boldsymbol{x}}\right].$$

This is centered log-Lorenz dominance of $\boldsymbol{y}$ over $\boldsymbol{x}$, a proportional equality criterion distinct from Lorenz dominance in income levels.

Despite their different economic interpretations, absolute and relative Dynamic Lorenz dominance share a common structure. In each case, a transition generates a rank-based effect profile, and the dominance relation compares cumulative centered effects at every lower-rank cutoff. In the absolute dimension, centering removes a uniform dollar change; in the relative dimension, it removes a uniform proportional change. On a fixed-total-income domain, absolute income changes already sum to zero, so the absolute criterion can equivalently be written using raw changes. This common centered-effect structure underlies both the elementary characterizations and the rank-weighted interpretation developed below. It is expressed using $K \in \{A, R\}$, with $K = A$ for the absolute dimension and $K = R$ for the relative dimension.

The paper provides axiomatic characterizations of both dominance relations. Symmetry makes the evaluation anonymous in each period. An elementary Dynamic Pigou–Dalton Principle requires that a progressive enhancement of the relevant rank-based K-effect profile make a transition strictly more equalizing, independently of the baseline on which the enhancement is realized. Centered Baseline Independence complements this directional requirement by imposing indifference between transitions with the same centered rank-based K-effect profile. Elementary progressivity thus governs strict comparisons generated by progressive enhancements, while Centered Baseline Independence governs the zero-enhancement boundary at which the centered profiles coincide. A finite-realization argument shows that every Dynamic Lorenz comparison can be generated through a feasible sequence of elementary enhancements. Together with Symmetry, these requirements are equivalent to consistency with the corresponding Dynamic Lorenz dominance relation, providing an elementary characterization of the cumulative-rank criterion in both the absolute and relative dimensions without assigning additional normative weight to the initial distribution.

The dominance relations are ordinal and do not require the selection of a particular inequality index. Their index-free interpretation is described in Section 2.8. As shown in a separate paper (Carbonell-Nicolau, 2026b), at fixed population size, K-Dynamic Lorenz dominance is equivalent to unanimity among the linear dynamic indices generated by weakly decreasing rank weights, while strict dominance yields strict agreement among those generated by strictly decreasing weights. Carbonell-Nicolau (2026b) also studies the broader measurement problem for baseline-independent indices, including the static inequality rankings that admit such dynamic extensions, the full class of admissible extensions, and the subclass that preserves Dynamic Lorenz comparisons.

The main analysis fixes population size but permits total income to vary. Accordingly, both dominance relations are defined directly through centered rank-based effect profiles. The analysis then extends the framework to varying population sizes by placing endpoint distributions on a common population-share grid and comparing the resulting cumulative transition effects. The formulation is invariant to replication and independent of the common refinement used to represent populations of different sizes.

The framework is related to several strands of literature. The income mobility literature typically retains the intertemporal association between individuals' initial and final positions (e.g., Shorrocks, 1978; Fields and Ok, 1999). Cowell (1985) axiomatizes cardinal indices of distributional change defined on matched initial and final incomes, with mobility and horizontal inequity as leading applications. Plotnick (1982) develops an ordinal criterion, analogous to the Lorenz criterion, for comparing the horizontal inequity of redistributions that share an initial ranking and a final distribution but differ in the assignment of final incomes to initial positions. Jenkins and Van Kerm (2006) use matched longitudinal incomes to decompose changes in generalized-Gini inequality into a progressivity component and a reranking component. These approaches retain individual-level

information across periods or give normative significance to the initial ranking. The present framework instead imposes anonymity separately in each period. It compares transitions between marginal distributions and therefore abstracts from reranking and exchange mobility.

A related literature evaluates how income growth is distributed across income levels or rank positions. Ravallion and Chen (2003) introduce the growth-incidence curve, which records income growth at each quantile, and use it to construct a measure of pro-poor growth linked to changes in the Watts poverty index. Son (2004) instead proposes a poverty-growth curve based on the growth of mean income among the bottom p proportion of the population and relates it to generalized-Lorenz dominance and reductions in poverty. Ray and Genicot (2023) conceptualize upward mobility as pro-poor growth. Their central Growth Progressivity axiom requires that transferring instantaneous growth rates from a richer to a poorer baseline income increase upward mobility. This operation is closely related to the elementary progressive enhancement used in the present paper, especially in the relative dimension, where transition effects are log-income changes.

The frameworks differ both in their objects and in their treatment of the baseline. Ray and Genicot derive a cardinal mobility index that assigns greater weight to growth at lower baseline incomes. Dynamic Lorenz dominance instead provides an ordinal comparison of cumulative rank-effect profiles. Its elementary progressivity requirement evaluates progressive enhancements independently of the baselines on which they are realized, while Centered Baseline Independence identifies transitions whose centered effect profiles coincide. Thus, Ray and Genicot measure upward mobility through a baseline-sensitive cardinal index, whereas the present framework identifies baseline-neutral transition comparisons without selecting a particular index.

The closest precedent for the cumulative dominance relation is Bossert and Dutta (2019). They require a measure of welfare change to be expressible as final social welfare minus initial social welfare, a property they call Welfare Difference Compatibility. Their second-order dominance condition compares rank-based income changes cumulatively: one transition dominates another if, at every lower-rank cutoff, its cumulative income change is at least as large. They show that this condition is equivalent to unanimous welfare-change comparisons within the generalized-Gini class. When all endpoint distributions have the same mean, their cumulative condition coincides formally with absolute Dynamic Lorenz dominance.

The approaches differ, however, in object, scope, and axiomatic characterization. Bossert and Dutta evaluate cardinal differences in social welfare, so changes in aggregate income remain part of welfare change. The present paper instead ranks the inequality-reducing content of transitions directly and removes the dimension-specific uniform component before comparing transition effects. Moreover, the relative dimension uses centered log effects and yields a proportional, scale-neutral dominance criterion with no counterpart in their raw-difference framework. Finally, Bossert and Dutta characterize cardinal generalized-Gini measures of welfare change, whereas the characterization here concerns an ordinal dynamic inequality pre-order generated by Symmetry, an elementary Dynamic Pigou–Dalton Principle, and Centered Baseline Independence.

The relative branch also connects to static results on logarithmic majorization and proportional transfers. Aboudi and Thon (2003) compare individual income distributions rather than transitions. They characterize weak supermajorization of ordered log incomes through multiplicative analogues of T-transforms, coordinatewise increases, and unanimity over additive welfare functions more concave than the logarithm. Their order retains the geometric-mean level and is therefore the logarithmic analogue of generalized-Lorenz welfare dominance. The centered criterion developed here removes that level component and is instead the logarithmic analogue of Lorenz inequality

dominance.

Fleurbaey and Michel (2001) and Aboudi and Thon (2003) provide complementary static foundations for proportional transfer comparisons. The former characterize the restrictions that proportional transfer principles impose on additive social welfare functions, while the latter develop the corresponding majorization orders and multiplicative transformations. Neither compares pairs of endpoint distributions through centered rank-based log-effect profiles, which are the object of the relative dynamic theory developed here.

The static transfer-sensitivity analysis of Shorrocks and Foster (1987) concerns a related but distinct refinement of Pigou–Dalton reasoning: it favors otherwise comparable physical transfers occurring at lower income levels, whereas the present dynamic analysis compares the rank reach of enhancements to transition-effect profiles. This distinction is discussed in Section 2.4.

The rest of the paper is organized as follows. Section 2 develops the fixed-population framework, establishes the elementary characterizations, provides their rank-weighted interpretation, and derives the static restrictions. Section 3 extends the analysis to varying population sizes. Section 4 concludes.

2 Fixed-population Dynamic Lorenz dominance

This section develops the absolute and relative criteria for a fixed population size $n \geq 1$, allowing total income to vary across distributions and periods, and establishes their elementary characterizations. When $n = 1$, every centered effect profile is zero, so all Dynamic Lorenz comparisons reduce to indifference and the elementary progressivity axiom is vacuous.

2.1 Basic setup and Symmetry

An *income distribution* for a population of size n is a vector of nonnegative incomes

$$\hat{\boldsymbol{x}} = (\hat{x}_1, \ldots, \hat{x}_n) \in \mathbb{R}^n_+$$

with strictly positive mean. Let $\Omega_{(n)}$ denote the space of all such unordered distributions, and let $\Omega^+_{(n)}$ denote its restriction to strictly positive incomes. A *transition* is a pair of endpoint distributions. The absolute transition space is $\Omega_{(n)} \times \Omega_{(n)}$, while the relative transition space is $\Omega^+_{(n)} \times \Omega^+_{(n)}$.

For $K \in \{A, R\}$, a *dynamic inequality pre-order* $\succcurlyeq$ is a reflexive and transitive binary relation on the corresponding transition space. We interpret

$$(\hat{\boldsymbol{x}}', \hat{\boldsymbol{y}}') \succcurlyeq (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$$

as saying that the reduction in inequality from $\hat{\boldsymbol{x}}'$ to $\hat{\boldsymbol{y}}'$ is weakly larger than the reduction from $\hat{\boldsymbol{x}}$ to $\hat{\boldsymbol{y}}$. The null transition $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{x}})$ represents no distributional change. Comparing $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$ with this null benchmark recovers a static comparison between $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$.

Because individuals differ only by income, the evaluation should be invariant under independent relabelings within the two periods. For a permutation π of $\{1, \ldots, n\}$, write $\hat{\boldsymbol{x}}^\pi = (\hat{x}_{\pi(1)}, \ldots, \hat{x}_{\pi(n)})$ for the corresponding relabeling of $\hat{\boldsymbol{x}}$.

Symmetry (S). For any permutations π and σ of $\{1, \ldots, n\}$, $(\hat{\boldsymbol{x}}^\pi, \hat{\boldsymbol{y}}^\sigma) \sim (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$.

Under S, distributions may be represented by their order statistics. For $\hat{\boldsymbol{x}} \in \Omega_{(n)}$, let $\boldsymbol{x} = (x_1, \ldots, x_n)$ denote its rank-ordered counterpart, where $x_1 \leq \cdots \leq x_n$. Rank refers to a position in

this ordered vector, not to a persistent individual identity. Let $\mathscr{D}_{(n)}$ and $\mathscr{D}^{+}_{(n)}$ denote the ordered counterparts of $\Omega_{(n)}$ and $\Omega^{+}_{(n)}$.

2.2 Transition-effect profiles

Because endpoint distributions are ordered independently, a physical transfer between particular individuals need not generate a two-coordinate change in the rank-based effect profile. If the transfer induces reranking, the incomes occupying several rank positions may change. Consequently, the same nominal transfer can generate different rank-effect profiles at different baselines. The analysis evaluates transitions through the changes they generate across rank positions, rather than through the particular physical transfers producing those changes.

For an ordered transition $T = (\boldsymbol{x}, \boldsymbol{y})$, let

$$\tau_i^A(T) = y_i - x_i$$

be its *absolute effect* at rank i. On the strictly positive domain, let

$$\tau_i^R(T) = \ln y_i - \ln x_i = \ln(y_i/x_i)$$

be its *relative effect*. Write $\boldsymbol{\tau}^K(T)$ for the corresponding transition-effect vector and

$$\bar{\tau}^K(T) = \frac{1}{n}\sum_{j=1}^{n} \tau_j^K(T), \qquad \widetilde{\boldsymbol{\tau}}^K(T) = \boldsymbol{\tau}^K(T) - \bar{\tau}^K(T)\mathbf{1}$$

for its mean and centered profile.

The centered effects admit dimension-specific interpretations. In the absolute dimension,

$$\widetilde{\tau}_i^A(T) = (y_i - \mu_{\boldsymbol{y}}) - (x_i - \mu_{\boldsymbol{x}}),$$

where $\mu_z = n^{-1}\sum_j z_j$. Thus, centering removes a common dollar change. In the relative dimension,

$$\widetilde{\tau}_i^R(T) = \ln\left(\frac{y_i}{\mu_{\boldsymbol{y}}^G}\right) - \ln\left(\frac{x_i}{\mu_{\boldsymbol{x}}^G}\right),$$

where $\mu_z^G = (\prod_j z_j)^{1/n}$. Thus, centering removes a common proportional change.

Remark 1. Log differences are used rather than ordinary percentage changes because they treat reciprocal proportional changes symmetrically: doubling and halving have equal magnitudes and opposite signs. They also add over time, since successive multiplicative changes become additive in log-income space (Törnqvist et al., 1985). The same log-income geometry underlies the static majorization order and multiplicative transformations studied by Aboudi and Thon (2003). Log income is also used as a socioeconomic coordinate by Esteban and Ray (1994), who motivate it through the symmetry of proportional income differences. Their application concerns the distribution of polarization within a population; the present analysis uses log differences to compare proportional changes across rank positions in income transitions.

2.3 Dynamic Lorenz dominance

Dynamic Lorenz dominance compares the cumulative centered effects accruing to lower ranks.

Definition 1 (K-Dynamic Lorenz dominance)**.** Let $T' = (\hat{\boldsymbol{x}}', \hat{\boldsymbol{y}}')$ and $T = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$ be transitions in the relevant fixed-population domain. Let $(\boldsymbol{x}', \boldsymbol{y}')$ and $(\boldsymbol{x}, \boldsymbol{y})$ denote their corresponding rank-ordered

representations, obtained by ordering each endpoint independently. For $K \in \{A, R\}$, transition T' *K-dynamically Lorenz dominates* T, written $T' \succcurlyeq^K_{DL} T$, if

$$\sum_{i=1}^{k} \tilde{\tau}_i^K(T') \geq \sum_{i=1}^{k} \tilde{\tau}_i^K(T), \qquad \text{for every } k \in \{1, \ldots, n\}.$$

Because every centered profile sums to zero, the condition at $k = n$ always holds with equality. Equivalently, if

$$d_k^K(T) = \sum_{i=1}^{k} \tau_i^K(T),$$

then the criterion requires

$$d_k^K(T') - d_k^K(T) \geq \frac{k}{n}\left[d_n^K(T') - d_n^K(T)\right], \qquad \text{for every } k.$$

This alternative expression makes the role of centering explicit. The right-hand term adjusts for the difference between the two transitions' average effects. After this adjustment, the criterion compares only how their effects are distributed across ranks.

On a fixed-total-income subdomain, $\bar{\tau}^A(T) = 0$ for every transition, so absolute Dynamic Lorenz dominance reduces to a cumulative comparison of raw income changes. In the relative dimension, centering removes the change in the log geometric mean, so the criterion applies directly even when the endpoint geometric means differ.

2.3.1 Worked transition comparisons

Dynamic Lorenz dominance compares active transitions, not only an active transition with a null benchmark. We first revisit the two transitions introduced in the Introduction:

$$A : (60, 140) \to (80, 120) \qquad \text{and} \qquad B : (10, 190) \to (20, 180).$$

Both are non-null transitions and begin from different baselines. Table 1 reports their effects, the mean removed by centering, the resulting centered profiles, and their cumulative centered effects. Because each centered profile sums to zero, only the first cutoff is nontrivial in this two-rank example.

Table 1: Absolute and relative evaluations of transitions A and B

Transition and dimension	Effect profile	Mean	Centered profile	Cumulative effect at $k = 1$
A, absolute	$(20, -20)$	0	$(20, -20)$	20
B, absolute	$(10, -10)$	0	$(10, -10)$	10
A, relative	$(0.2877, -0.1542)$	0.0668	$(0.2209, -0.2209)$	0.2209
B, relative	$(0.6931, -0.0541)$	0.3195	$(0.3736, -0.3736)$	0.3736

In the absolute dimension, A has the larger cumulative centered effect, so $A \succ^A_{DL} B$. In the relative dimension, the ranking is reversed: $B \succ^R_{DL} A$. Thus, A produces the stronger redistribution in income units, whereas B produces the stronger redistribution in proportional positions.

Dynamic Lorenz dominance is incomplete, as is ordinary Lorenz dominance. To illustrate, consider two further non-null transitions from different baselines:

$$C : (66, 79, 212) \to (74, 85, 218) \quad \text{and} \quad D : (26, 46, 75) \to (27, 50, 73).$$

All four endpoint distributions are ordered. Table 2 reports the absolute and relative calculations. The final cumulative centered effect is zero in every row and is therefore omitted.

Table 2: Transitions incomparable in both dimensions

Transition and dimension	Effect profile	Mean	Centered profile	Cumulative effects at $k = 1, 2$
C, absolute	$(8, 6, 6)$	6.6667	$(1.3333, -0.6667, -0.6667)$	$(1.3333, 0.6667)$
D, absolute	$(1, 4, -2)$	1	$(0, 3, -3)$	$(0, 3)$
C, relative	$(0.1144, 0.0732, 0.0279)$	0.0718	$(0.0426, 0.0014, -0.0439)$	$(0.0426, 0.0439)$
D, relative	$(0.0377, 0.0834, -0.0270)$	0.0314	$(0.0064, 0.0520, -0.0584)$	$(0.0064, 0.0584)$

In the absolute comparison, C has the larger cumulative centered effect at the first cutoff, while D has the larger one at the second cutoff. Hence,

$$C \not\succcurlyeq^{A}_{DL} D \qquad \text{and} \qquad D \not\succcurlyeq^{A}_{DL} C.$$

The relative cumulative profiles also cross: C is larger at the first cutoff and D at the second. Thus,

$$C \not\succcurlyeq^{R}_{DL} D \qquad \text{and} \qquad D \not\succcurlyeq^{R}_{DL} C.$$

Table 2 gives two non-null transitions whose cumulative profiles cross in both dimensions. Neither transition dynamically Lorenz dominates the other, so the criterion leaves the pair incomparable. Resolving the comparison would require additional normative restrictions.

Lemma 1. *For each $K \in \{A, R\}$, $\succcurlyeq^{K}_{DL}$ is a reflexive and transitive relation satisfying* S.

Proof. For $\boldsymbol{z} \in \mathbb{R}^n$, define

$$L_k(\boldsymbol{z}) = \sum_{i=1}^{k} z_i - \frac{k}{n} \sum_{i=1}^{n} z_i.$$

Then $T' \succcurlyeq^{K}_{DL} T$ if and only if $L_k(\boldsymbol{\tau}^K(T') - \boldsymbol{\tau}^K(T)) \geq 0$ for every k. Reflexivity is immediate, and transitivity follows from the linearity of L_k. Independent endpoint permutations leave the rank-ordered endpoints, and hence all L_k, unchanged. ∎

2.4 Elementary progressive enhancements

The primitive progressivity axiom concerns one enhancement at a time. Let

$$\boldsymbol{\delta}(r, d; a) = a(\boldsymbol{e}_r - \boldsymbol{e}_d), \qquad a > 0, \quad r < d.$$

This vector adds effect at a lower rank and removes the same amount at a higher rank. Its cumulative effect is

$$\sum_{i=1}^{k} \delta_i(r, d; a) = \begin{cases} 0, & k < r, \\ a, & r \leq k < d, \\ 0, & k \geq d. \end{cases}$$

Elementary K-Dynamic Pigou–Dalton Principle (E-K-DPDP). For $K \in \{A, R\}$, if two ordered transitions T' and T satisfy

$$\widetilde{\boldsymbol{\tau}}^K(T') = \widetilde{\boldsymbol{\tau}}^K(T) + a(\boldsymbol{e}_r - \boldsymbol{e}_d)$$

for some $a > 0$ and $r < d$, then $T' \succ T$.

The axiom compares valid endpoint transitions. It does not require the enhancement to be executed as a physical transfer on either endpoint. The realization results below show that finite sequences of such elementary profile improvements can be represented by feasible transitions on a common auxiliary baseline.

In both dimensions, elementary enhancements can be compared by their effect intensity and rank reach. For $\boldsymbol{\delta} = a(\boldsymbol{e}_r - \boldsymbol{e}_d)$ and $\boldsymbol{\delta}' = a'(\boldsymbol{e}_{r'} - \boldsymbol{e}_{d'})$, $\boldsymbol{\delta}'$ cumulatively dominates $\boldsymbol{\delta}$ whenever $r' \le r < d \le d'$ and $a' \ge a$. The first condition says that the rank interval of $\boldsymbol{\delta}'$ contains that of $\boldsymbol{\delta}$; the second says that its effect intensity is weakly larger. In the absolute dimension, intensity is measured in income units; in the relative dimension, it is measured in log-change units.

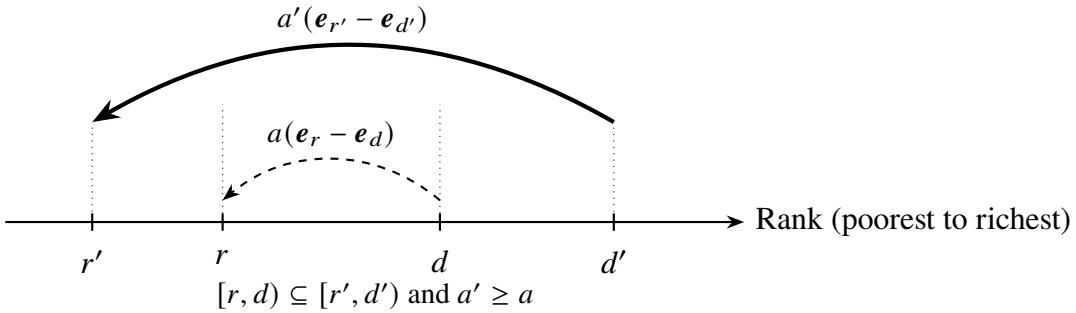


Figure 1: Nested rank reach of elementary enhancements. Under the displayed nesting and intensity conditions, the upper enhancement has a weakly larger cumulative effect at every lower-rank cutoff. In the absolute dimension, intensity is measured in income units; in the relative dimension, it is measured in log-change units.

The nested rank reach illustrated in Figure 1 differs from the transfer-sensitivity property of Shorrocks and Foster (1987). Their analysis refines the static Pigou–Dalton comparison by considering the income-level location of otherwise comparable progressive transfers. The present comparison concerns a different object: holding effect intensity fixed, one elementary enhancement is stronger when its rank interval contains that of the other. Transfer sensitivity concerns the income-level location of static transfers, whereas Figure 1 concerns the nested rank reach of enhancements to transition-effect profiles.

2.5 Centered Baseline Independence and dimension-specific neutrality

Elementary progressivity determines the direction of a single improvement, but does not identify transitions that generate the same distributively relevant effect profile.

Centered K-Baseline Independence (K-CBI). For $K \in \{A, R\}$, if two ordered transitions T and U satisfy

$$\widetilde{\boldsymbol{\tau}}^K(T) = \widetilde{\boldsymbol{\tau}}^K(U),$$

then $T \sim U$.

The axiom makes the centered rank-effect profile sufficient for evaluating a transition. It abstracts from baseline income levels and baseline inequality once that profile has been fixed. It also treats the uniform component of a transition's effects as distributively neutral: adding the same dollar effect at every rank leaves the centered absolute profile unchanged, while adding the same log effect at every rank leaves the centered relative profile unchanged. The latter corresponds to a common proportional growth factor.

For completeness, these neutrality implications can be expressed directly.

Dynamic Translation Invariance (DTI). For all admissible $\alpha, \beta \in \mathbb{R}$,

$$(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}) \sim (\hat{\boldsymbol{x}} + \alpha\mathbf{1}, \hat{\boldsymbol{y}} + \beta\mathbf{1}).$$

Dynamic Scale Invariance (DSI). For all $\alpha, \beta > 0$,

$$(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}) \sim (\beta\hat{\boldsymbol{x}}, \alpha\hat{\boldsymbol{y}}).$$

Proposition 1. *Under* S, *A*-CBI *implies* DTI, *and R*-CBI *implies* DSI.

Proof. Consider first the absolute dimension. Let $T = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$, and let $(\boldsymbol{x}, \boldsymbol{y})$ be its rank-ordered representation. For admissible $\alpha, \beta \in \mathbb{R}$, the rank-ordered representation of

$$(\hat{\boldsymbol{x}} + \alpha\mathbf{1}, \hat{\boldsymbol{y}} + \beta\mathbf{1})$$

is

$$(\boldsymbol{x} + \alpha\mathbf{1}, \boldsymbol{y} + \beta\mathbf{1}).$$

Its absolute-effect profile differs from that of $(\boldsymbol{x}, \boldsymbol{y})$ by $(\beta - \alpha)\mathbf{1}$, which disappears on centering. Therefore, *A*-CBI gives

$$(\boldsymbol{x}, \boldsymbol{y}) \sim (\boldsymbol{x} + \alpha\mathbf{1}, \boldsymbol{y} + \beta\mathbf{1}).$$

By S, each arbitrary transition is indifferent to its rank-ordered representation. Transitivity consequently gives

$$(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}) \sim (\hat{\boldsymbol{x}} + \alpha\mathbf{1}, \hat{\boldsymbol{y}} + \beta\mathbf{1}),$$

which is DTI.

The relative argument is analogous. Positive rescaling preserves the ordering of each endpoint. Rescaling the two endpoints changes the relative-effect profile by a uniform vector, which disappears on centering. *R*-CBI gives indifference between the corresponding rank-ordered transitions, and S extends that indifference to the original, potentially unordered transitions. Hence DSI holds. ■

The relevant invariance differs across the two dimensions. In the absolute dimension, adding a common dollar amount to either endpoint changes every rank's absolute effect by the same amount, which centering removes. Rescaling an endpoint is not neutral because it generally changes dollar differences across ranks. In the relative dimension, multiplying either endpoint by a common positive factor changes every rank's log effect by the same amount, which centering again removes. Adding a common dollar amount is not neutral in this dimension because it generally changes proportional income differences. Thus, together with Symmetry, Centered Baseline Independence implies Dynamic Translation Invariance in the absolute dimension and Dynamic Scale Invariance in the relative dimension.

2.6 Elementary foundations

The next results turn the cumulative criterion into a finite sequence of feasible elementary comparisons.

Lemma 2. *Let $\boldsymbol{q} \in \mathbb{R}^n$ satisfy $\sum_{i=1}^n q_i = 0$, and define*

$$Q_k = \sum_{i=1}^{k} q_i, \qquad k \in \{1, \dots, n-1\}.$$

Then

$$\boldsymbol{q} = \sum_{k=1}^{n-1} Q_k(\boldsymbol{e}_k - \boldsymbol{e}_{k+1}).$$

Consequently, $Q_k \geq 0$ for every k if and only if $\boldsymbol{q}$ is a finite sum of adjacent progressive enhancements, after zero terms are omitted.

Proof. Expanding the right-hand side gives

$$\sum_{k=1}^{n-1} Q_k(\boldsymbol{e}_k - \boldsymbol{e}_{k+1}) = Q_1\boldsymbol{e}_1 + \sum_{i=2}^{n-1}(Q_i - Q_{i-1})\boldsymbol{e}_i - Q_{n-1}\boldsymbol{e}_n.$$

By the definition of the cumulative sums,

$$Q_1 = q_1 \qquad \text{and} \qquad Q_i - Q_{i-1} = q_i \quad \text{for } i \in \{2, \dots, n-1\}.$$

Moreover, because $\sum_{i=1}^n q_i = 0$,

$$-Q_{n-1} = -\sum_{i=1}^{n-1} q_i = q_n.$$

Hence every coordinate on the right-hand side equals the corresponding coordinate of $\boldsymbol{q}$, proving

$$\boldsymbol{q} = \sum_{k=1}^{n-1} Q_k(\boldsymbol{e}_k - \boldsymbol{e}_{k+1}).$$

If $Q_k \geq 0$ for every k, each nonzero term in this decomposition is an adjacent progressive enhancement. Conversely, if $\boldsymbol{q}$ is a finite sum of adjacent progressive enhancements, then each enhancement has nonnegative cumulative sums at every lower-rank cutoff. By addition, $Q_k \geq 0$ for every k. ∎

The following two lemmas convert a finite algebraic path into feasible transitions sharing one auxiliary baseline.

Lemma 3. *Let $\boldsymbol{z}^0, \dots, \boldsymbol{z}^M \in \mathbb{R}^n$ be a finite family of zero-sum vectors. There exists a strictly increasing vector $\boldsymbol{b} \in \mathbb{R}^n_{++}$ such that*

$$\boldsymbol{b}, \boldsymbol{b} + \boldsymbol{z}^m \in \mathscr{D}_{(n)}, \qquad \text{for every } m \in \{0, \dots, M\}.$$

Hence every $\boldsymbol{z}^m$ is the centered absolute-effect profile of the admissible transition $(\boldsymbol{b}, \boldsymbol{b} + \boldsymbol{z}^m)$.

Proof. For $i \in \{1, \dots, n-1\}$, choose gaps g_i satisfying

$$g_i > \max\left\{0, \max_{0 \leq m \leq M}(z_i^m - z_{i+1}^m)\right\}.$$

Construct $\boldsymbol{b}$ recursively from $b_{i+1} - b_i = g_i$. Then

$$(b_{i+1} + z_{i+1}^m) - (b_i + z_i^m) > 0$$

for every i and m. Finally add a sufficiently large common constant to all coordinates of $\boldsymbol{b}$ so that $\boldsymbol{b}$ and every $\boldsymbol{b} + \boldsymbol{z}^m$ are strictly positive. This translation preserves all adjacent gaps. Since $\sum_i z_i^m = 0$, the absolute effects of $(\boldsymbol{b}, \boldsymbol{b} + \boldsymbol{z}^m)$ are already centered and equal $\boldsymbol{z}^m$. ■

Lemma 4. *Let $\boldsymbol{z}^0, \ldots, \boldsymbol{z}^M \in \mathbb{R}^n$ be a finite family of zero-sum vectors. There exists a strictly increasing log-baseline $\boldsymbol{\beta} \in \mathbb{R}^n$ such that*

$$e^{\boldsymbol{\beta}}, e^{\boldsymbol{\beta}+\boldsymbol{z}^m} \in \mathscr{D}_{(n)}^{+}, \qquad \text{for every } m \in \{0, \ldots, M\},$$

where exponentiation is componentwise. Hence every $\boldsymbol{z}^m$ is the centered relative-effect profile of the admissible transition $(e^{\boldsymbol{\beta}}, e^{\boldsymbol{\beta}+\boldsymbol{z}^m})$.

Proof. For each $i \in \{1, \ldots, n-1\}$, choose the adjacent gap of $\boldsymbol{\beta}$ so that

$$\beta_{i+1} - \beta_i > \max\left\{0, \max_{0 \le m \le M} (z_i^m - z_{i+1}^m)\right\}.$$

These strictly positive gaps make $\boldsymbol{\beta}$ strictly increasing. They also imply $(\beta_{i+1} + z_{i+1}^m) - (\beta_i + z_i^m) > 0$ for every i and m, so every $\boldsymbol{\beta} + \boldsymbol{z}^m$ is strictly increasing. Exponentiation preserves order and produces strictly positive income vectors. The relative-effect profile of $(e^{\boldsymbol{\beta}}, e^{\boldsymbol{\beta}+\boldsymbol{z}^m})$ is exactly $\boldsymbol{z}^m$. Since $\sum_i z_i^m = 0$, its mean is zero, so its centered relative-effect profile is also $\boldsymbol{z}^m$. ■

Proposition 2 (feasible elementary path)**.** *If $T' \succcurlyeq_{DL}^{K} T$, then there are transitions $U^0, \ldots, U^M$ in the relevant fixed-population domain such that*

$$\widetilde{\boldsymbol{\tau}}^K(U^0) = \widetilde{\boldsymbol{\tau}}^K(T), \qquad \widetilde{\boldsymbol{\tau}}^K(U^M) = \widetilde{\boldsymbol{\tau}}^K(T'),$$

and every nontrivial step satisfies

$$\widetilde{\boldsymbol{\tau}}^K(U^m) = \widetilde{\boldsymbol{\tau}}^K(U^{m-1}) + a_m(\boldsymbol{e}_{r_m} - \boldsymbol{e}_{d_m})$$

for some $a_m > 0$ and $r_m < d_m$. All U^m may be chosen to share one ordered baseline. If the original dominance is strict, the path contains at least one nontrivial step.

Proof. Let

$$\boldsymbol{z}^0 = \widetilde{\boldsymbol{\tau}}^K(T) \qquad \text{and} \qquad \boldsymbol{q} = \widetilde{\boldsymbol{\tau}}^K(T') - \widetilde{\boldsymbol{\tau}}^K(T).$$

Because both centered effect profiles have zero sum,

$$\sum_{i=1}^{n} q_i = 0.$$

For $k \in \{1, \ldots, n-1\}$, define

$$Q_k = \sum_{i=1}^{k} q_i.$$

The assumption $T' \succcurlyeq_{DL}^{K} T$ is exactly the requirement that

$$Q_k \ge 0, \qquad \text{for every } k \in \{1, \ldots, n-1\}.$$

By Lemma 2,

$$\boldsymbol{q} = \sum_{k=1}^{n-1} Q_k(\boldsymbol{e}_k - \boldsymbol{e}_{k+1}).$$

Omit the terms for which $Q_k = 0$, and list the remaining terms as

$$a_m(\boldsymbol{e}_{r_m} - \boldsymbol{e}_{r_m+1}), \qquad m \in \{1, \dots, M\},$$

where every $a_m > 0$. Define the intermediate profiles recursively by

$$\boldsymbol{z}^m = \boldsymbol{z}^{m-1} + a_m(\boldsymbol{e}_{r_m} - \boldsymbol{e}_{r_m+1}), \qquad m \in \{1, \dots, M\}.$$

Each $\boldsymbol{z}^m$ has zero sum, and

$$\boldsymbol{z}^M = \boldsymbol{z}^0 + \boldsymbol{q} = \widetilde{\boldsymbol{\tau}}^K(T').$$

Thus, each successive profile is obtained from the preceding one by a single adjacent progressive enhancement.

When $K = A$, apply Lemma 3 to the finite family $\boldsymbol{z}^0, \dots, \boldsymbol{z}^M$. When $K = R$, apply Lemma 4. In either case, there exist admissible transitions

$$U^0, \dots, U^M$$

on one common auxiliary baseline such that

$$\widetilde{\boldsymbol{\tau}}^K(U^m) = \boldsymbol{z}^m, \qquad \text{for every } m \in \{0, \dots, M\}.$$

Hence, for every $m \in \{1, \dots, M\}$,

$$\widetilde{\boldsymbol{\tau}}^K(U^m) = \widetilde{\boldsymbol{\tau}}^K(U^{m-1}) + a_m(\boldsymbol{e}_{r_m} - \boldsymbol{e}_{r_m+1}),$$

so U^m is obtained from U^{m-1} by one feasible elementary progressive enhancement.

If $T' \succ^K_{DL} T$, at least one Dynamic Lorenz inequality is strict. Equivalently, $Q_k > 0$ for some k, so the decomposition contains at least one nonzero enhancement and $M \geq 1$. If all Dynamic Lorenz inequalities hold with equality, then $\boldsymbol{q} = \boldsymbol{0}$, and the construction reduces to the single profile $\boldsymbol{z}^0 = \widetilde{\boldsymbol{\tau}}^K(T') = \widetilde{\boldsymbol{\tau}}^K(T)$. ■

2.7 Elementary characterizations

A dynamic inequality pre-order $\succcurlyeq$ is *K-DL-consistent* if, for any two transitions T' and T in the relevant domain,

$$T' \succcurlyeq^K_{DL} T \Rightarrow T' \succcurlyeq T \quad \text{and} \quad T' \succ^K_{DL} T \Rightarrow T' \succ T.$$

For unordered transitions, Dynamic Lorenz dominance is evaluated through the rank-ordered representations of their endpoints. The preceding results provide the common architecture for the two characterizations. Lemma 2 expresses every cumulative dominance gap as a finite sequence of elementary progressive enhancements, while Lemma 3 and Lemma 4 place all intermediate profiles on one feasible auxiliary baseline. The two dimensions differ only in this realization step: absolute profiles are realized in income levels, whereas relative profiles are realized in log-income levels. The next two theorems apply this common argument separately to the absolute and relative criteria.

Theorem 1 (elementary characterization of absolute Dynamic Lorenz dominance)**.** *A dynamic inequality pre-order on $\Omega_{(n)} \times \Omega_{(n)}$ satisfies* S*,* E-*A*-DPDP*, and* *A*-CBI *if and only if it is A-DL-consistent.*

Proof. Suppose first that $\succcurlyeq$ is A-DL-consistent. If two transitions differ only by independent endpoint permutations, their rank-ordered counterparts coincide, so they dominate each other under $\succcurlyeq^A_{DL}$. Consistency gives indifference and hence S. If two ordered transitions have identical centered absolute-effect profiles, they again dominate each other, so consistency gives *A*-CBI.

Finally, one elementary progressive enhancement has weakly nonnegative cumulative sums and a strictly positive cumulative sum at every cutoff between its recipient and donor ranks, implying strict A-DL dominance. Consistency gives E-A-DPDP.

Conversely, suppose that S, E-A-DPDP, and A-CBI hold, and let $T' \succcurlyeq^{A}_{DL} T$. Work first with the rank-ordered counterparts. By Proposition 2, there is a feasible path $U^0, \dots, U^M$ on one ordered baseline such that

$$\widetilde{\tau}^{A}(U^0) = \widetilde{\tau}^{A}(T) \qquad \text{and} \qquad \widetilde{\tau}^{A}(U^M) = \widetilde{\tau}^{A}(T'),$$

and every nontrivial step is one elementary progressive enhancement. By A-CBI,

$$T \sim U^0 \qquad \text{and} \qquad U^M \sim T'.$$

If $M = 0$, the endpoint profiles coincide, and A-CBI gives $T' \sim T$. If $M \geq 1$, E-A-DPDP gives $U^m \succ U^{m-1}$ at every step, and transitivity gives $U^M \succ U^0$. In either case, $T' \succcurlyeq T$. If the original A-DL comparison is strict, Proposition 2 ensures $M \geq 1$, so $T' \succ T$. S transfers these conclusions from the ordered representatives to the original unordered transitions. Thus, $\succcurlyeq$ is A-DL-consistent. ■

Theorem 2 (elementary characterization of relative Dynamic Lorenz dominance)**.** *A dynamic inequality pre-order on $\Omega^{+}_{(n)} \times \Omega^{+}_{(n)}$ satisfies* S, E-R-DPDP, *and* R-CBI *if and only if it is R-DL-consistent.*

Proof. Suppose first that $\succcurlyeq$ is R-DL-consistent. The argument is analogous to the necessity part of the proof of Theorem 1, with centered relative-effect profiles in place of centered absolute-effect profiles. Thus, $\succcurlyeq$ satisfies S, E-R-DPDP, and R-CBI.

Conversely, suppose that S, E-R-DPDP, and R-CBI hold, and let $T' \succcurlyeq^{R}_{DL} T$. By Proposition 2, the centered log-effect profiles can be connected through feasible transitions $U^0, \dots, U^M$ sharing one log-income baseline. R-CBI gives

$$T \sim U^0 \qquad \text{and} \qquad U^M \sim T'.$$

If $M = 0$, the endpoint profiles coincide, and R-CBI gives $T' \sim T$. If $M \geq 1$, E-R-DPDP strictly ranks each elementary step, and transitivity gives $U^M \succ U^0$. Hence $T' \succcurlyeq T$ for every weak R-DL comparison. Strict R-DL dominance guarantees at least one nontrivial step, so it gives $T' \succ T$. S extends the conclusion from the rank-ordered representatives to the original unordered transitions. Thus, $\succcurlyeq$ is R-DL-consistent. ■

2.8 Connection to rank-weighted dynamic indices

Dynamic Lorenz dominance admits a dual interpretation through linear rank-weighted dynamic indices. For $K \in \{A, R\}$, define the centered endpoint coordinates by

$$q^{A}_{i}(z) = z_i - \mu_z \qquad \text{and} \qquad q^{R}_{i}(z) = \ln z_i - \overline{\ln z},$$

on the corresponding income domains. Given a rank-weight vector $\boldsymbol{w} = (w_1, \dots, w_n)$, let

$$I^{K}_{\boldsymbol{w}}(z) = \sum_{i=1}^{n} w_i q^{K}_{i}(z)$$

and define the associated linear dynamic index by

$$J^{K}_{\boldsymbol{w}}(\boldsymbol{x}, \boldsymbol{y}) = I^{K}_{\boldsymbol{w}}(\boldsymbol{y}) - I^{K}_{\boldsymbol{w}}(\boldsymbol{x}).$$

Equivalently,

$$J_w^K(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{n} w_i \widetilde{\tau}_i^K(\boldsymbol{x}, \boldsymbol{y}).$$

In the absolute dimension, this construction is related to the generalized-Gini measures of welfare change studied by Bossert and Dutta (2019). Their indices take differences of rank-weighted income levels and therefore retain changes in mean income. The centered coordinates used here remove the change in mean income and isolate the distributional component of the transition. The relative construction applies the same rank-weighted logic to centered log incomes.

As shown by Carbonell-Nicolau (2026b), at fixed population size, one transition K-dynamically Lorenz dominates another if and only if every index J_w^K generated by a weakly decreasing rank-weight vector, $w_1 \geq \cdots \geq w_n$, assigns it a weakly larger value. If the Dynamic Lorenz comparison is strict, every index generated by a strictly decreasing rank-weight vector assigns the dominating transition a strictly larger value. Thus, unanimity within the linear decreasing-rank-weight family recovers the Dynamic Lorenz dominance relation. The same paper characterizes the broader class of baseline-independent dynamic indices and identifies those that preserve Dynamic Lorenz comparisons.

2.9 Static and fixed-mean special cases

The variable-mean analysis contains the familiar fixed-mean and static comparisons as restrictions.

On a fixed-total-income subdomain,

$$\bar{\tau}^A(T) = 0,$$

so centered absolute effects coincide with raw income changes. On this subdomain, absolute Dynamic Lorenz dominance is simply the cumulative comparison of raw income changes, and the elementary characterization in Theorem 1 applies without modification.

For any dynamic inequality pre-order $\succcurlyeq$, define its *induced static relation* $\succcurlyeq^*$ by

$$\hat{\boldsymbol{y}} \succcurlyeq^* \hat{\boldsymbol{x}} \Leftrightarrow (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}) \succcurlyeq (\hat{\boldsymbol{x}}, \hat{\boldsymbol{x}}).$$

Thus, an active transition is compared with the null transition at its own baseline.

The induced static relation need not be transitive. Accordingly, the static characterization below is stated directly for a static inequality pre-order—that is, a reflexive and transitive relation on the positive income domain—rather than for the relation induced by an arbitrary dynamic pre-order.

2.9.1 The absolute static restriction

Against the null transition, absolute Dynamic Lorenz dominance requires

$$\sum_{i=1}^{k} (y_i - \mu_{\boldsymbol{y}}) \geq \sum_{i=1}^{k} (x_i - \mu_{\boldsymbol{x}}), \qquad \text{for every } k.$$

On a common-mean domain, this is ordinary Lorenz dominance. Moreover, an elementary absolute enhancement of a null transition raises the income at one lower rank and reduces the income at one higher rank by the same amount. In this sense, the variable-mean dynamic theory recovers the classical Pigou–Dalton operation and the Lorenz criterion as static special cases.

2.9.2 Centered log-Lorenz dominance

Against the null transition, relative Dynamic Lorenz dominance compares mean-centered log-income profiles.

Definition 2 (centered log-Lorenz dominance)**.** For positive ordered distributions $\boldsymbol{x}, \boldsymbol{y} \in \mathscr{D}^{+}_{(n)}$, write $\boldsymbol{y} \succcurlyeq_{CL} \boldsymbol{x}$ if

$$\sum_{i=1}^{k} \ln\left(\frac{y_i}{\mu_{\boldsymbol{y}}^{G}}\right) \geq \sum_{i=1}^{k} \ln\left(\frac{x_i}{\mu_{\boldsymbol{x}}^{G}}\right), \qquad \text{for every } k \in \{1, \ldots, n\}.$$

For unordered distributions, the comparison is made through their ordered counterparts.

For a positive distribution $\boldsymbol{x}$, define

$$\ell(\boldsymbol{x}) = \ln \boldsymbol{x} - \overline{\ln \boldsymbol{x}}\mathbf{1} = \ln\left(\frac{\boldsymbol{x}}{\mu_{\boldsymbol{x}}^{G}}\right).$$

Then $\boldsymbol{y} \succcurlyeq_{CL} \boldsymbol{x}$ if and only if $\ell(\boldsymbol{y})$ Lorenz-majorizes $\ell(\boldsymbol{x})$. In other words, centered log-Lorenz dominance applies ordinary majorization after each distribution has been normalized by its geometric mean.

The elementary static operation is correspondingly formulated in log-income space.

Centered Proportional Transfer Principle (CPTP). A static inequality pre-order $\succcurlyeq^{*}$ on $\Omega^{+}_{(n)}$ satisfies CPTP if, for any ordered $\boldsymbol{x}, \boldsymbol{y} \in \mathscr{D}^{+}_{(n)}$,

$$\ell(\boldsymbol{y}) = \ell(\boldsymbol{x}) + a(\boldsymbol{e}_r - \boldsymbol{e}_d)$$

for some $a > 0$ and $r < d$ implies $\boldsymbol{y} \succ^{*} \boldsymbol{x}$.

Equivalently, the raw log-income gap has the form

$$\ln \boldsymbol{y} - \ln \boldsymbol{x} = a(\boldsymbol{e}_r - \boldsymbol{e}_d) + c\mathbf{1}$$

for some $c \in \mathbb{R}$. The first term is a zero-sum progressive transfer in log-income space; the second is a common proportional rescaling and is neutral for relative inequality. If both distributions are required to lie on a common arithmetic-mean slice, the scalar c is the unique mean-restoring adjustment. On the present variable-mean domain, no such restoration is required.

Because the domain allows geometric means to vary, CPTP and S alone do not identify distributions that differ only by scale. The appropriate static neutrality condition is also required.

Scale Invariance (SI). For every $\hat{\boldsymbol{x}} \in \Omega^{+}_{(n)}$ and every $\lambda > 0$, $\lambda\hat{\boldsymbol{x}} \sim^{*} \hat{\boldsymbol{x}}$.

A static pre-order $\succcurlyeq^{*}$ is *centered-log-Lorenz consistent* if it preserves the weak and strict comparisons made by $\succcurlyeq_{CL}$.

Theorem 3 (static characterization of centered log-Lorenz dominance)**.** *A static inequality pre-order on $\Omega^{+}_{(n)}$ satisfies S, CPTP, and SI if and only if it is centered-log-Lorenz consistent.*

Proof. Suppose first that $\succcurlyeq^{*}$ is centered-log-Lorenz consistent. Permutations and positive rescalings leave ordered centered log profiles unchanged, so consistency in both directions gives S and SI. A CPTP step adds one progressive transfer to the centered log profile. Its cumulative sums are nonnegative and are strictly positive for $r \leq k < d$, so it gives strict centered log-Lorenz dominance. Consistency gives CPTP.

Conversely, suppose that $\succcurlyeq^*$ satisfies S, CPTP, and SI, and let $\boldsymbol{y} \succcurlyeq_{CL} \boldsymbol{x}$. Work with ordered representatives and write

$$\boldsymbol{u} = \ell(\boldsymbol{x}) \qquad \text{and} \qquad \boldsymbol{v} = \ell(\boldsymbol{y}).$$

Both vectors are ordered and have zero sum. Moreover, centered log-Lorenz dominance means that

$$\sum_{i=1}^{k} v_i \geq \sum_{i=1}^{k} u_i, \qquad \text{for every } k.$$

By the ordered Hardy–Littlewood–Pólya transfer theorem (Hardy et al., 1952; Marshall and Olkin, 1979; Marshall et al., 2011), there exists a finite sequence of ordered zero-sum vectors

$$\boldsymbol{u} = \boldsymbol{z}^0, \boldsymbol{z}^1, \ldots, \boldsymbol{z}^M = \boldsymbol{v}$$

such that every nontrivial step has the form

$$\boldsymbol{z}^m = \boldsymbol{z}^{m-1} + a_m(\boldsymbol{e}_{r_m} - \boldsymbol{e}_{d_m}),$$

where $a_m > 0$ and $r_m < d_m$.

For each m, let $\boldsymbol{w}^m = e^{\boldsymbol{z}^m}$, with exponentiation defined componentwise. Since $\boldsymbol{z}^m$ is ordered, $\boldsymbol{w}^m$ is a positive ordered distribution. Moreover, $\sum_i z_i^m = 0$ implies that $\mu^G_{\boldsymbol{w}^m} = 1$, and hence $\ell(\boldsymbol{w}^m) = \boldsymbol{z}^m$. Every nontrivial step from $\boldsymbol{w}^{m-1}$ to $\boldsymbol{w}^m$ is consequently a CPTP step. CPTP and transitivity give

$$e^{\boldsymbol{v}} \succcurlyeq^* e^{\boldsymbol{u}},$$

with strict preference if $\boldsymbol{y} \succ_{CL} \boldsymbol{x}$, since a strict comparison implies $\boldsymbol{v} \neq \boldsymbol{u}$ and hence at least one nontrivial step.

Finally,

$$\boldsymbol{x} = \mu^G_{\boldsymbol{x}} e^{\boldsymbol{u}} \qquad \text{and} \qquad \boldsymbol{y} = \mu^G_{\boldsymbol{y}} e^{\boldsymbol{v}}.$$

By SI,

$$\boldsymbol{x} \sim^* e^{\boldsymbol{u}} \qquad \text{and} \qquad \boldsymbol{y} \sim^* e^{\boldsymbol{v}}.$$

Combining these indifferences with the preceding comparison gives

$$\boldsymbol{y} \succcurlyeq^* \boldsymbol{x},$$

strictly whenever $\boldsymbol{y} \succ_{CL} \boldsymbol{x}$. S extends the conclusion from the ordered representatives to the original unordered distributions. ■

The additional SI axiom is essential on the variable-mean domain. If two distributions have the same centered log profile, they differ only by a positive scalar. Centered log-Lorenz dominance treats them as equivalent, but CPTP alone governs only strict progressive changes and does not impose this scale indifference.

Theorem 3 is related to the logarithmic majorization result of Aboudi and Thon (2003), but the two criteria differ in their treatment of scale. Aboudi and Thon compare cumulative ordered log incomes without centering and characterize the resulting order through multiplicative transformations and unanimity over a class of additive welfare functions. Their criterion retains differences in the geometric mean. Centered log-Lorenz dominance instead subtracts mean log income before applying majorization, so distributions that differ only by a common positive rescaling are equivalent.

2.9.3 Absolute and proportional judgments need not agree

Centered log-Lorenz dominance is not a refinement of ordinary Lorenz dominance. The two criteria evaluate different distributive geometries.

Consider

$$\boldsymbol{x} = (10, 50, 90) \qquad \text{and} \qquad \boldsymbol{y} = (15, 40, 95).$$

The distributions have the same mean. The cumulative differences in income levels, taking $\boldsymbol{y} - \boldsymbol{x}$, are

$$5 \qquad \text{and} \qquad -5$$

at the first and second cutoffs. Because these differences change sign, ordinary Lorenz dominance does not compare the distributions.

The difference between their centered log-income profiles is approximately

$$\ell(\boldsymbol{y}) - \ell(\boldsymbol{x}) = (0.3267, -0.3019, -0.0247).$$

Its cumulative sums are

$$0.3267 \qquad \text{and} \qquad 0.0247$$

at the first and second cutoffs. Both are positive, so $\boldsymbol{y} \succ_{CL} \boldsymbol{x}$. The movement from $\boldsymbol{x}$ to $\boldsymbol{y}$ is unambiguously equalizing in proportional positions even though it is not Lorenz-improving in income levels.

Now consider

$$\boldsymbol{z} = (10, 60, 80).$$

Distribution $\boldsymbol{z}$ is obtained from $\boldsymbol{x}$ by a progressive physical transfer from the richest to the middle rank, so $\boldsymbol{z}$ Lorenz dominates $\boldsymbol{x}$. The difference between their centered log-income profiles is approximately

$$\ell(\boldsymbol{z}) - \ell(\boldsymbol{x}) = (-0.0215, 0.1608, -0.1393).$$

Its cumulative sums are

$$-0.0215 \qquad \text{and} \qquad 0.1393$$

at the first and second cutoffs. Because these differences change sign, centered log-Lorenz dominance does not compare $\boldsymbol{z}$ and $\boldsymbol{x}$. A physical Pigou–Dalton improvement can thus remain unranked by the proportional criterion, just as an improvement in proportional positions can remain unranked by the level-income criterion.

2.10 Baseline neutrality and related transition comparisons

The elementary characterizations impose baseline neutrality through two complementary requirements. E-K-DPDP governs strict comparisons by ranking progressive enhancements independently of the baseline on which they are realized, while K-CBI imposes indifference when the centered effect profiles coincide. This joint restriction is substantive: in the Dynamic Lorenz comparisons characterized by the two axioms, realizing a centered effect profile at lower baseline incomes or against a more unequal initial distribution carries no additional normative weight.

The framework of Ray and Genicot (2023) illustrates a baseline-sensitive alternative in the relative dimension. Their Growth Progressivity axiom requires upward mobility to increase when, holding the baseline-income vector fixed, instantaneous growth rates are transferred from a richer to a poorer baseline income. Together with their auxiliary axioms, this requirement yields an index whose weights on growth rates depend on baseline incomes and decline as baseline income rises.

The contrast with the present framework involves both E-R-DPDP and R-CBI. E-R-DPDP ranks a given progressive enhancement independently of the baseline on which it is realized, while R-CBI imposes indifference when the centered relative-effect profiles coincide. Relaxing these baseline-neutrality requirements could lead to either a baseline-weighted theory, in which baseline income affects the value assigned to effects at particular ranks, or a baseline-dependent theory, in which the initial degree of inequality enters as an additional aggregate feature. Developing those alternatives lies beyond the present paper.

The absolute criterion also clarifies the relation to Bossert and Dutta (2019). Their welfare-change dominance condition compares cumulative raw income differences. It coincides with absolute Dynamic Lorenz dominance when the aggregate absolute-effect gap vanishes. When means vary, their condition retains aggregate income change, whereas the present criterion removes its uniform component by centering. The two approaches also differ in their axiomatic objects: Bossert and Dutta characterize cardinal welfare-change measures, whereas the present theorems characterize ordinal consistency with a dynamic inequality dominance relation. The relative centered-log geometry has no counterpart in their raw-difference framework.

3 Varying population sizes

We now allow the four endpoint distributions in a comparison to have different population sizes. Let

$$\Omega = \bigcup_{n\geq 1} \Omega_{(n)} \qquad \text{and} \qquad \Omega^{+} = \bigcup_{n\geq 1} \Omega^{+}_{(n)}.$$

The unrestricted transition spaces are $\Omega \times \Omega$ in the absolute dimension and $\Omega^{+} \times \Omega^{+}$ in the relative dimension. For a distribution $\hat{\boldsymbol{x}}$, let $n_{\hat{\boldsymbol{x}}}$ denote its population size.

3.1 Replication and common refinements

For an integer $m \geq 1$, the m-replica of $\hat{\boldsymbol{x}} = (\hat{x}_1, \ldots, \hat{x}_n)$ is

$$\hat{\boldsymbol{x}}^{\{m\}} = (\underbrace{\hat{x}_1, \ldots, \hat{x}_1}_{m \text{ times}}, \ldots, \underbrace{\hat{x}_n, \ldots, \hat{x}_n}_{m \text{ times}}).$$

Replication preserves the arithmetic and geometric means and repeats every centered rank coordinate on a finer population-share grid.

Dynamic Population Principle (DPP). For every transition $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$ and all integers $l, m \geq 1$,

$$(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}) \sim (\hat{\boldsymbol{x}}^{\{l\}}, \hat{\boldsymbol{y}}^{\{m\}}).$$

DPP permits the two periods to be replicated independently. This is natural because the framework compares anonymous endpoint distributions rather than tracking a fixed set of individuals across periods.

DPP also contains the usual static replication comparison. Starting from the null transition $(\boldsymbol{x}, \boldsymbol{x})$, replicate only the final endpoint by a positive integer ℓ. Then

$$(\boldsymbol{x}, \boldsymbol{x}^{\{\ell\}}) \sim (\boldsymbol{x}, \boldsymbol{x}).$$

Thus, when the static relation is induced from the unrestricted dynamic pre-order in the same way as in Section 2.9, $\boldsymbol{x}^{\{\ell\}} \sim^{*} \boldsymbol{x}$.

To compare two unrestricted transitions, choose a positive integer N divisible by the population sizes of all four endpoints. Their rank-ordered N-replicas form a *common refinement*. For an ordered distribution z whose population divides N, write

$$z^{[N]} = z^{\{N/n_z\}}.$$

For an unrestricted transition $T = (\hat{x}, \hat{y})$, define its centered K-effect profile on the N-grid by

$$\widetilde{\tau}^{K,[N]}(T) = \widetilde{\tau}^{K}(x^{[N]}, y^{[N]}).$$

Definition 3 (unrestricted K-Dynamic Lorenz dominance)**.** Let T' and T be unrestricted transitions, and let N be a common refinement size. For $K \in \{A, R\}$, write $T' \succcurlyeq^{K}_{DL} T$ if

$$\sum_{i=1}^{k} \widetilde{\tau}_i^{K,[N]}(T') \geq \sum_{i=1}^{k} \widetilde{\tau}_i^{K,[N]}(T), \qquad \text{for every } k \in \{1, \ldots, N\}.$$

The definition is independent of the selected common refinement. The next subsection first illustrates the construction and then establishes its refinement invariance, empirical-quantile representation, and basic order-theoretic properties.

3.1.1 The common-refinement construction and its properties

We begin with a population-changing example. Consider the two transitions

$$T : (25, 45) \to (30, 30, 60) \quad \text{and} \quad T' : (25, 35) \to (50, 55, 60).$$

In each transition, the population increases from two to three. To compare the transitions, place all four endpoint distributions on a common six-person grid. Replicating each initial distribution three times and each final distribution twice gives

$$x^{[6]} = (25, 25, 25, 45, 45, 45), \qquad y^{[6]} = (30, 30, 30, 30, 60, 60),$$

and

$$x'^{[6]} = (25, 25, 25, 35, 35, 35), \qquad y'^{[6]} = (50, 50, 55, 55, 60, 60).$$

Table 3 reports the absolute effects and their centered cumulative profiles on this common grid.

Table 3: A population-changing comparison on a common refinement

Transition	Absolute-effect profile	Mean effect	Centered effect profile	Cumulative centered effects at $k \in \{1, \ldots, 5\}$
T	$(5, 5, 5, -15, 15, 15)$	5	$(0, 0, 0, -20, 10, 10)$	$(0, 0, 0, -20, -10)$
T'	$(25, 25, 30, 20, 25, 25)$	25	$(0, 0, 5, -5, 0, 0)$	$(0, 0, 5, 0, 0)$

At every cutoff, the cumulative centered absolute effect of T' is weakly larger than that of T, with strict inequality at the third, fourth, and fifth cutoffs. Hence $T' \succ^{A}_{DL} T$. Notice that the comparison does not treat the larger average income change under T' as itself equalizing. Centering removes that uniform component. The ranking arises from how the residual absolute effects are distributed across population shares.

The same conclusion is obtained on any further common refinement. For example, on a twelve-person grid every coordinate in the six-person profiles is repeated twice. The cumulative gaps at the original grid points are doubled, while the gaps at the intervening points linearly interpolate

between them. No inequality changes sign. This illustrates the refinement invariance established next.

Lemma 5 (refinement invariance)**.** *If the inequalities in Definition 3 hold on one common refinement, they hold on every common refinement. Strict dominance is likewise independent of the refinement.*

Proof. We first compare a common refinement of size N with a further refinement of size qN, where q is a positive integer. If z is an ordered endpoint whose population divides N, then

$$z^{[qN]} = (z^{[N]})^{\{q\}}.$$

Thus, passing from the N-grid to the qN-grid repeats each coordinate of every endpoint q times. Replication preserves arithmetic means and mean log incomes. Therefore, it leaves the mean absolute or relative effect unchanged, so each coordinate of the centered effect profile is repeated q times.

Let A_j and B_j denote the cumulative centered effects of T' and T, respectively, through rank j on the N-grid, and set $A_0 = B_0 = 0$. Write

$$D_j = A_j - B_j.$$

The inequalities in Definition 3 on the N-grid are exactly

$$D_j \geq 0, \qquad \text{for every } j \in \{1, \ldots, N\}.$$

Consider rank

$$k = q(j-1) + s$$

on the qN-grid, where $j \in \{1, \ldots, N\}$ and $s \in \{1, \ldots, q\}$. At that rank, the first $j-1$ coordinates of the N-grid have each been counted q times, while coordinate j has been counted s times. Hence the cumulative gap on the qN-grid is

$$qD_{j-1} + s(D_j - D_{j-1}) = (q-s)D_{j-1} + sD_j.$$

If all D_j are nonnegative, every refined-grid gap is nonnegative. Conversely, at rank qj, the refined-grid gap is qD_j, so nonnegativity on the qN-grid implies nonnegativity on the N-grid. The weak dominance inequalities thus hold on one grid if and only if they hold on the other.

This equivalence also preserves strictness. If $D_j > 0$, the refined-grid gap at rank qj is $qD_j > 0$. Conversely, if a refined-grid gap

$$(q-s)D_{j-1} + sD_j$$

is positive while the weak inequalities hold, then $D_{j-1} > 0$ or $D_j > 0$.

For arbitrary common refinement sizes N and N', let

$$L = \operatorname{lcm}(N, N').$$

Since L is a further common refinement of both grids, the preceding equivalence shows that the weak and strict comparisons on the N-grid, the L-grid, and the N'-grid coincide. ∎

Remark 2 (empirical-quantile formulation)**.** The common-refinement construction has an equivalent empirical-quantile interpretation. Each ordered distribution z defines the step quantile function

$$Q_z(p) = z_i, \qquad \text{for } p \in \left(\frac{i-1}{n_z}, \frac{i}{n_z}\right].$$

Replication leaves this function unchanged while representing it on a finer population-share grid. The centered absolute and relative effect profiles can accordingly be viewed as step functions on

$(0, 1]$, obtained from differences between the corresponding endpoint quantile functions, in levels or logs, after centering. Unrestricted K-Dynamic Lorenz dominance is equivalent to comparing the integrals of the two centered effect functions over $(0, p]$. Because these functions are constant between successive population shares at which one of the four endpoint quantile functions changes value, it is enough to compare their cumulative integrals at those finitely many population shares.

Lemma 6. *For each $K \in \{A, R\}$, unrestricted K-Dynamic Lorenz dominance is reflexive and transitive and satisfies* S *and* DPP.

Proof. Reflexivity follows immediately from the reflexivity of the fixed-population relation.

For transitivity, suppose that

$$T^1 \succcurlyeq^K_{DL} T^2 \qquad \text{and} \qquad T^2 \succcurlyeq^K_{DL} T^3.$$

Choose a common refinement size N divisible by the population sizes of all six endpoint distributions. By Lemma 5, both assumed comparisons hold on this same N-grid. Hence

$$T^{1,[N]} \succcurlyeq^K_{DL} T^{2,[N]} \qquad \text{and} \qquad T^{2,[N]} \succcurlyeq^K_{DL} T^{3,[N]}.$$

The fixed-population relation is transitive by Lemma 1, so $T^{1,[N]} \succcurlyeq^K_{DL} T^{3,[N]}$. The definition of unrestricted dominance then gives $T^1 \succcurlyeq^K_{DL} T^3$.

S follows because independent endpoint permutations leave the rank-ordered replicas unchanged.

For DPP, let

$$T = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}) \qquad \text{and} \qquad T^{\ell,m} = (\hat{\boldsymbol{x}}^{\{\ell\}}, \hat{\boldsymbol{y}}^{\{m\}}),$$

where $\ell, m \geq 1$. Choose a common refinement size N divisible by both $\ell n_{\hat{\boldsymbol{x}}}$ and $m n_{\hat{\boldsymbol{y}}}$. On the N-grid,

$$\left(\boldsymbol{x}^{\{\ell\}}\right)^{[N]} = \boldsymbol{x}^{[N]} \qquad \text{and} \qquad \left(\boldsymbol{y}^{\{m\}}\right)^{[N]} = \boldsymbol{y}^{[N]}.$$

Indeed, refining the ℓ-replica of $\boldsymbol{x}$ to population N repeats each original coordinate a total of

$$\ell \frac{N}{\ell n_{\boldsymbol{x}}} = \frac{N}{n_{\boldsymbol{x}}}$$

times, exactly as in $\boldsymbol{x}^{[N]}$; the same argument applies to $\boldsymbol{y}$. Thus T and $T^{\ell,m}$ have identical endpoint representations, and hence identical centered effect profiles, on the common N-grid. They dominate each other under unrestricted K-Dynamic Lorenz dominance, so $T \sim^K_{DL} T^{\ell,m}$. This is DPP. ∎

3.2 Within-grid axioms and cross-grid replication

The elementary progressivity and baseline-independence axioms continue to govern comparisons within each fixed-population subdomain. Population replication has a separate role: it connects transitions represented on different population grids.

For each $n \geq 1$, consider only comparisons between transitions whose four endpoints all have population size n. Say that a dynamic inequality pre-order on the unrestricted transition space satisfies E-K-DPDP *on every fixed-population subdomain* if E-K-DPDP holds for all such comparisons, for every $n \geq 1$. Likewise, it satisfies K-CBI *on every fixed-population subdomain* if K-CBI holds for all such comparisons, for every $n \geq 1$.

These requirements compare transitions whose endpoints already lie on one common grid. They do not identify an n-person transition with a replicated representation on a different grid. That cross-grid indifference is supplied by DPP.

3.3 Unrestricted elementary characterizations

A dynamic inequality pre-order on an unrestricted transition space is *unrestricted K-DL-consistent* if it preserves the weak and strict comparisons in Definition 3.

Theorem 4 (unrestricted absolute characterization)**.** *A dynamic inequality pre-order on $\Omega \times \Omega$ is unrestricted A-DL-consistent if and only if it satisfies:*

(i) S*;*
(ii) E-*A*-DPDP *on every fixed-population subdomain;*
(iii) *A*-CBI *on every fixed-population subdomain; and*
(iv) DPP.

Theorem 5 (unrestricted relative characterization)**.** *A dynamic inequality pre-order on $\Omega^+ \times \Omega^+$ is unrestricted R-DL-consistent if and only if it satisfies:*

(i) S*;*
(ii) E-*R*-DPDP *on every fixed-population subdomain;*
(iii) *R*-CBI *on every fixed-population subdomain; and*
(iv) DPP.

Proof of Theorem 4 and Theorem 5. Fix $K \in \{A, R\}$. Suppose first that $\succcurlyeq$ is unrestricted K-DL-consistent. Independent endpoint permutations leave every common-refinement profile unchanged, so mutual unrestricted K-DL dominance and consistency imply S. On any fixed-population subdomain, unrestricted K-Dynamic Lorenz dominance coincides with the relation in Definition 1. The necessity arguments in Theorem 1 and Theorem 2 then imply E-K-DPDP and K-CBI on that subdomain. Finally, a transition and any independently replicated version have identical representations on a common refinement. They dominate each other under unrestricted K-Dynamic Lorenz dominance, so consistency implies DPP.

Conversely, suppose that the four conditions hold, and let $T' \succcurlyeq^{K}_{DL} T$. Choose a common refinement size N for the four endpoint distributions, and let $T^{[N]}$ and $T'^{[N]}$ denote the resulting N-person transitions. By DPP, $T \sim T^{[N]}$ and $T' \sim T'^{[N]}$. By the definition of unrestricted dominance, $T'^{[N]} \succcurlyeq^{K}_{DL} T^{[N]}$.

When comparisons are confined to the N-person domain, $\succcurlyeq$ satisfies S, E-K-DPDP, and K-CBI. The applicable fixed-population characterization—Theorem 1 when $K = A$ and Theorem 2 when $K = R$—gives $T'^{[N]} \succcurlyeq T^{[N]}$.

If the unrestricted Dynamic Lorenz comparison is strict, Lemma 5 ensures that the comparison between the N-person representatives is also strict. The applicable fixed-population theorem then gives $T'^{[N]} \succ T^{[N]}$.

Finally, DPP gives $T' \sim T'^{[N]}$ and $T^{[N]} \sim T$. In the weak case,

$$T' \succcurlyeq T'^{[N]} \succcurlyeq T^{[N]} \succcurlyeq T,$$

so $T' \succcurlyeq T$. In the strict case,

$$T' \sim T'^{[N]} \succ T^{[N]} \sim T,$$

so $T' \succ T$. Thus, $\succcurlyeq$ is unrestricted K-DL-consistent. ∎

4 Concluding remarks

This paper develops an ordinal, index-free framework for comparing inequality changes across income transitions. Its central object is the centered rank-based effect profile of a transition. The absolute dimension evaluates income changes after removing their uniform dollar component, while the relative dimension evaluates log-income changes after removing their uniform proportional component. Dynamic Lorenz dominance compares the cumulative centered effects of two transitions at every lower-rank cutoff.

On fixed-population domains with unrestricted total income, Symmetry, an elementary K-Dynamic Pigou–Dalton Principle, and Centered K-Baseline Independence characterize the absolute and relative dominance relations. The elementary principle evaluates one progressive enhancement at a time. Adjacent decomposition and simultaneous realization show that every Dynamic Lorenz comparison can be generated through a feasible finite sequence of such enhancements. The cumulative criterion thus admits an elementary characterization.

The static restrictions further clarify the content of the two dimensions. On a common-mean domain, absolute Dynamic Lorenz dominance against a null transition reduces to ordinary Lorenz dominance. Relative Dynamic Lorenz dominance against a null transition yields centered log-Lorenz dominance, which applies ordinary majorization to geometric-mean-normalized incomes. The resulting criterion is scale neutral and distinct from ordinary Lorenz dominance in income levels. On the variable-mean domain, its static characterization requires Symmetry, the Centered Proportional Transfer Principle, and Scale Invariance.

The analysis also accommodates varying population sizes. Common replication places all endpoint distributions on one population grid. Refinement invariance makes the resulting dominance comparison independent of the chosen grid, while the Dynamic Population Principle links the original transitions to their replicated representations. The unrestricted characterization combines the elementary axioms applied within each fixed-population grid with this separate cross-grid replication principle.

Two extensions remain open. First, the baseline-neutrality requirements could be relaxed in two distinct ways. The present framework ranks progressive enhancements independently of the baselines on which they are realized and treats transitions with identical centered effect profiles alike. A baseline-weighted theory could allow baseline income to affect the value of transition effects at particular ranks. A baseline-dependent theory could instead allow an aggregate feature of the initial distribution, such as its degree of inequality, to enter the evaluation of the transition. The first approach modifies the rank-specific valuation of effects, whereas the second adds a feature of the baseline distribution as a whole.

Second, the population extension follows the classical replication principle and abstracts from absolute demographic scale. Replication preserves relative population positions and the cumulative shares used by Dynamic Lorenz dominance, but it may alter normatively relevant features tied to the number of individuals represented at particular parts of the distribution. Carbonell-Nicolau (2026a) develops population-sensitive criteria for static inequality comparisons by combining Lorenz dominance with the income shares held by fixed headcounts at the top and bottom of the distribution. Extending such demographic sensitivity to comparisons between income transitions remains an open problem.

A Axiom independence for the fixed-population characterizations

Fix $K \in \{A, R\}$ and a population size $n \geq 2$. On every such fixed-population domain, the three axioms in the corresponding characterization, Theorem 1 for $K = A$ and Theorem 2 for $K = R$, are logically independent. The restriction $n \geq 2$ is necessary because elementary progressive enhancements require two distinct ranks; when $n = 1$, every centered effect profile is zero. For each axiom, we construct a dynamic inequality pre-order satisfying the other two but failing K-DL-consistency.

Let $T = (\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$ denote an unordered transition, and let $T^\circ = (\boldsymbol{x}, \boldsymbol{y})$ denote its rank-ordered counterpart. For the appendix only, write

$$\hat{\tau}_i^A(T) = \hat{y}_i - \hat{x}_i \qquad \text{and} \qquad \hat{\tau}_i^R(T) = \ln \hat{y}_i - \ln \hat{x}_i$$

for the effects attached to the original labels.

Dropping S. For an unordered transition T, define the label-sensitive statistic

$$h_K(T) = \hat{\tau}_1^K(T) - \hat{\tau}_n^K(T).$$

Define $\succcurlyeq_{\neg S}$ lexicographically by

$$T' \succcurlyeq_{\neg S} T \Leftrightarrow \begin{cases} T'^\circ \succ_{DL}^K T^\circ, \\ \text{or} \\ T'^\circ \sim_{DL}^K T^\circ \text{ and } h_K(T') \geq h_K(T). \end{cases}$$

Pairs that are incomparable under K-Dynamic Lorenz dominance remain incomparable. This relation is reflexive and transitive because it lexicographically refines each K-DL-indifference class by a real-valued statistic. It satisfies E-K-DPDP because an elementary progressive enhancement gives strict K-DL dominance before the tie-breaker is reached. It also satisfies K-CBI on the ordered subspace. Indeed, if two ordered transitions T and U have the same centered profile, then their raw effect profiles differ by a uniform vector:

$$\boldsymbol{\tau}^K(T) = \boldsymbol{\tau}^K(U) + c\mathbf{1}$$

for some $c \in \mathbb{R}$. Hence

$$h_K(T) = \tau_1^K(T) - \tau_n^K(T) = \tau_1^K(U) - \tau_n^K(U) = h_K(U).$$

The transitions are K-DL-indifferent, and the label-sensitive tie-breaker does not distinguish them, so $T \sim_{\neg S} U$.

The relation fails S. Choose a transition T for which $h_K(T) \neq 0$, and apply the same permutation interchanging labels 1 and n to both endpoints, producing a transition T' with the same rank-ordered representation. Hence $T' \sim_{DL}^K T$. The permutation reverses the label-sensitive statistic, so $h_K(T') = -h_K(T)$. Because $h_K(T) \neq 0$, the relation $\succcurlyeq_{\neg S}$ strictly ranks T and T'. Thus, it fails S and does not preserve their K-DL indifference, so it is not K-DL-consistent.

Dropping E-K-DPDP. Define the reverse Dynamic Lorenz relation by

$$T' \succcurlyeq_{\neg\text{E-DPDP}} T \Leftrightarrow \sum_{i=1}^{k} \widetilde{\tau}_i^K(T') \leq \sum_{i=1}^{k} \widetilde{\tau}_i^K(T), \text{ for every } k.$$

It is reflexive and transitive, satisfies S, and treats identical centered profiles as indifferent, so it satisfies K-CBI. It reverses every strict K-Dynamic Lorenz comparison, including every elementary progressive enhancement. Hence, it fails E-K-DPDP and is not K-DL-consistent.

Dropping K-CBI. Let $G(\boldsymbol{x})$ denote the Gini coefficient of the ordered baseline. Define $\succcurlyeq_{\neg\text{CBI}}$ by

$$T' \succcurlyeq_{\neg\text{CBI}} T \Leftrightarrow \begin{cases} T'^{\circ} \succ_{DL}^{K} T^{\circ}, \\ \text{or} \\ T'^{\circ} \sim_{DL}^{K} T^{\circ} \text{ and } G(\boldsymbol{x}') \leq G(\boldsymbol{x}). \end{cases}$$

K-DL-incomparable pairs remain incomparable. This is a reflexive and transitive lexicographic refinement of K-Dynamic Lorenz dominance. It satisfies S because both the ordered Dynamic Lorenz relation and the baseline Gini coefficient are permutation invariant. It satisfies E-K-DPDP because an elementary progressive enhancement gives strict K-DL dominance before the baseline tie-breaker is reached.

The relation fails K-CBI. Choose two ordered baselines $\boldsymbol{x}$ and $\boldsymbol{u}$ with $G(\boldsymbol{u}) > G(\boldsymbol{x})$. Their null transitions have the same zero centered K-effect profile and are thus K-DL-indifferent, while

$$(\boldsymbol{x}, \boldsymbol{x}) \succ_{\neg\text{CBI}} (\boldsymbol{u}, \boldsymbol{u}).$$

Thus, for every fixed-population domain with $n \geq 2$, the three axioms S, E-K-DPDP, and K-CBI are logically independent.

Data availability statement

No data were used or generated in this theoretical study. No data archive or replication package is applicable.

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